

THE ROLE OF CHEMICAL BONDING AND HYDRATION STATE ON THERMAL EXPANSION OF A ZINC-BASED METAL-ORGANIC FRAMEWORK



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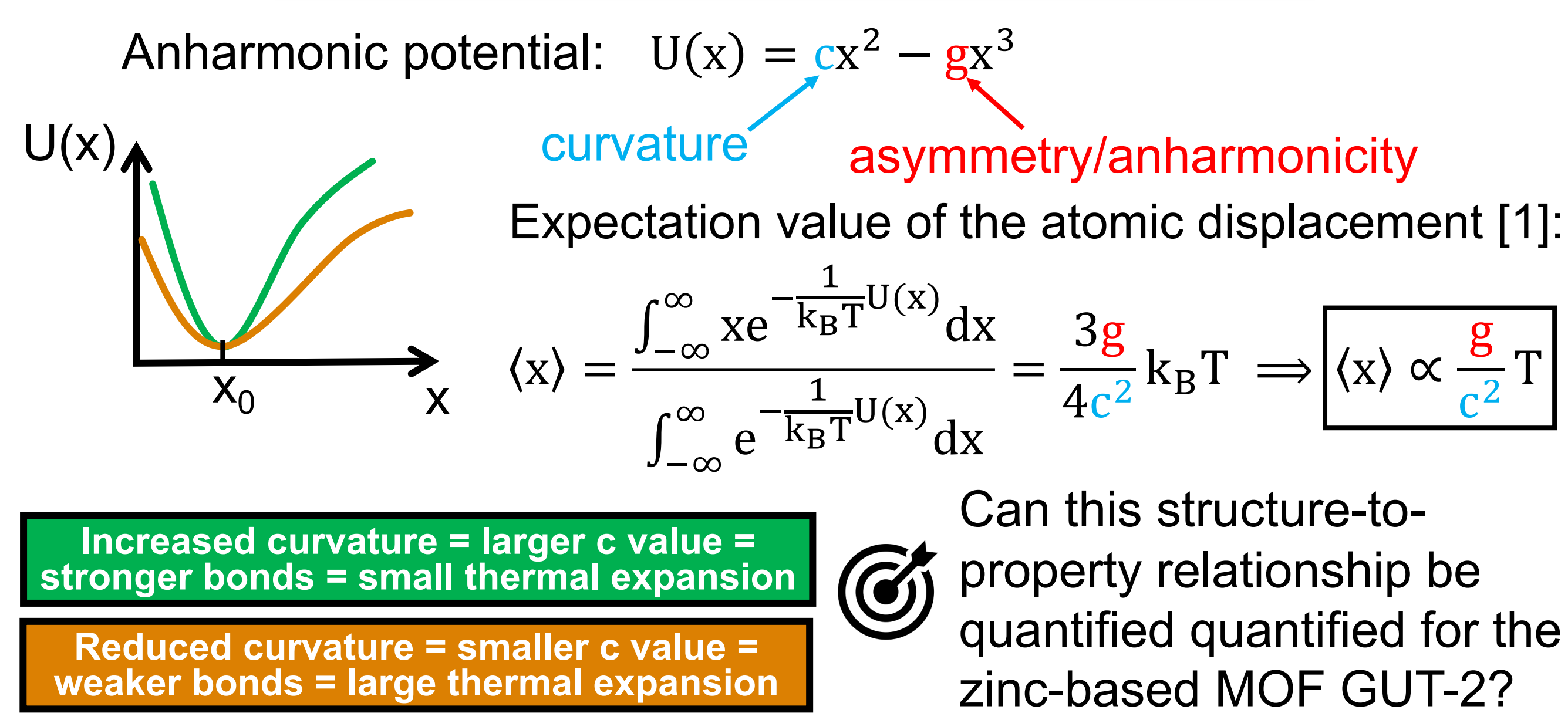
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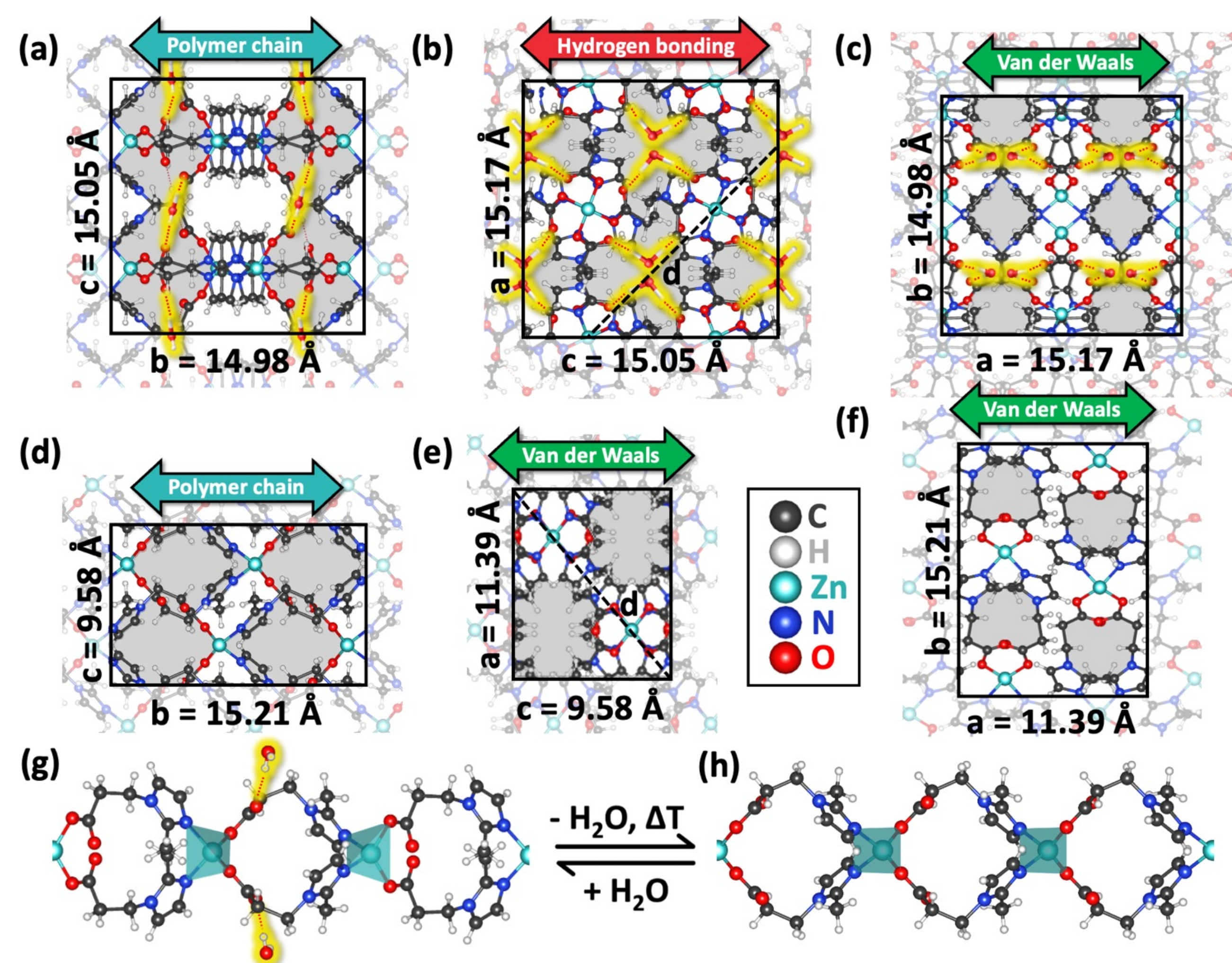


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INTRODUCTION: HOW CHEMICAL BONDS IMPACT THERMAL EXPANSION



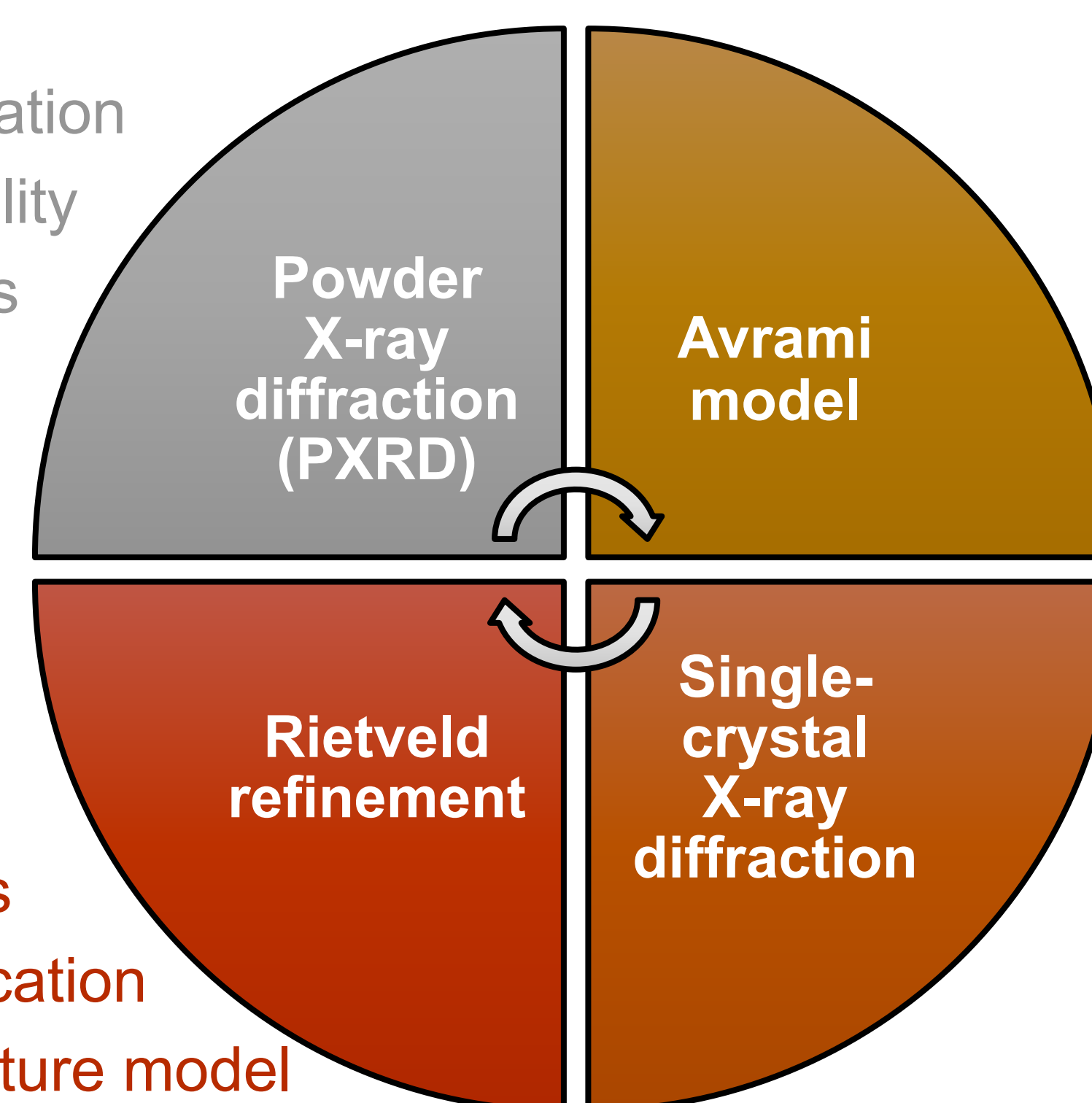
HYDRATED AND ACTIVATED CRYSTAL STRUCTURES OF GUT-2



EXPERIMENTAL METHODS AND ANALYSIS

- Phase identification
- Structural stability
- Crystallite sizes
- Transformation
- Framework integrity

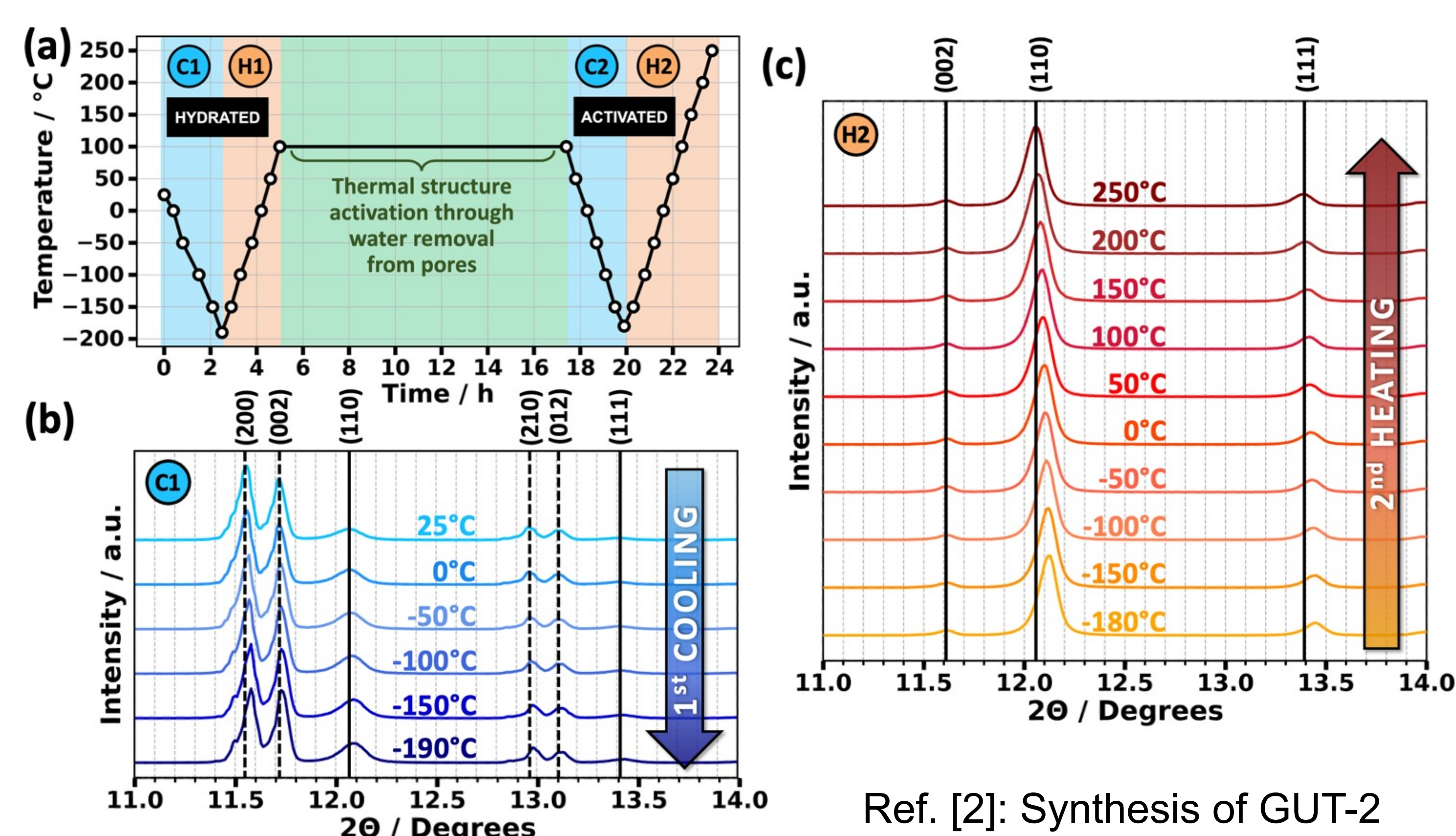
- Temperature-dependent cell parameters
- Phase quantification
- Improved structure model



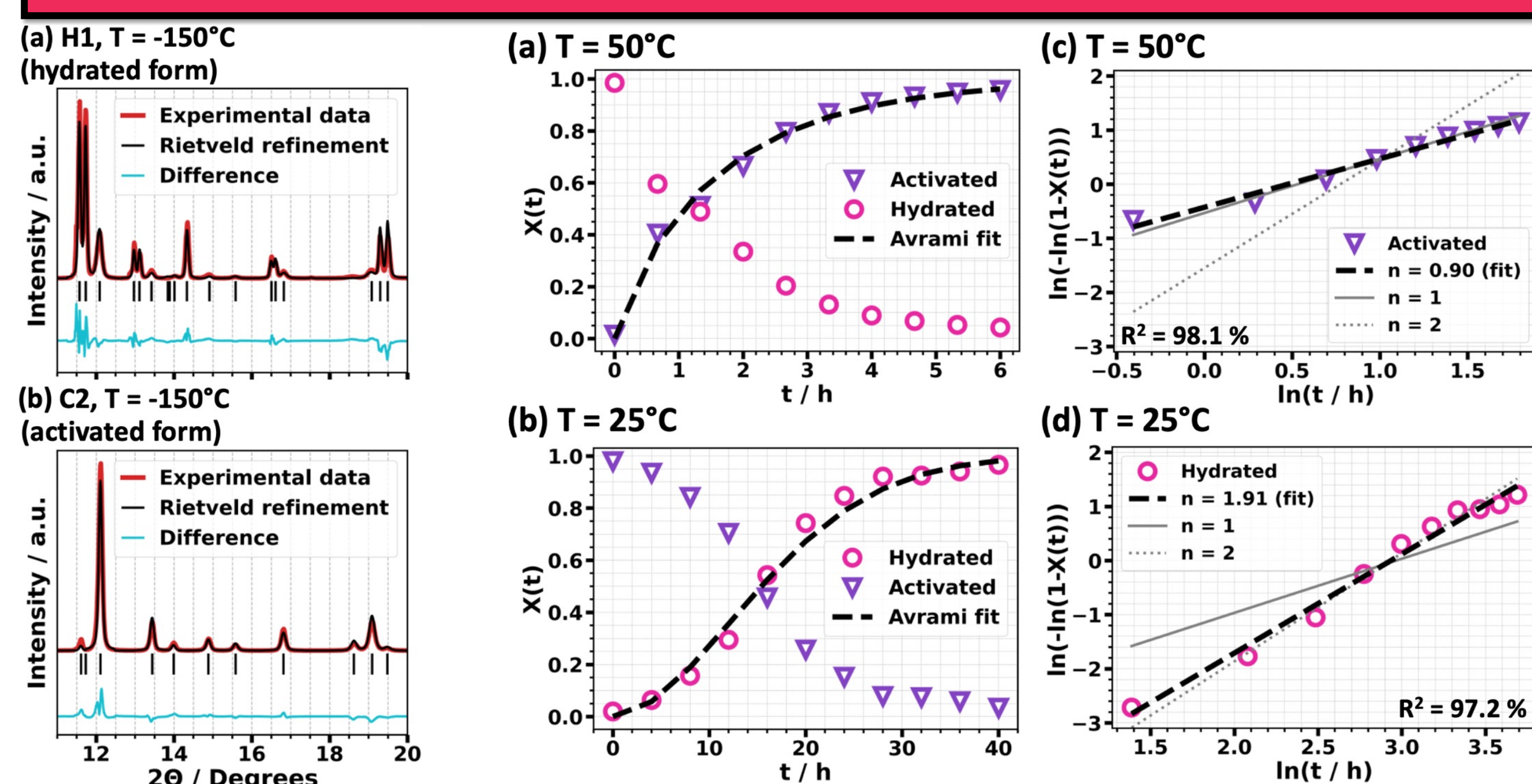
- Hydration and rehydration kinetics
- Dimensionality of the process
- Growth mechanism

- Atomic positions
- Unit cell lengths
- Space group
- H₂O localization
- Disorder, twinning

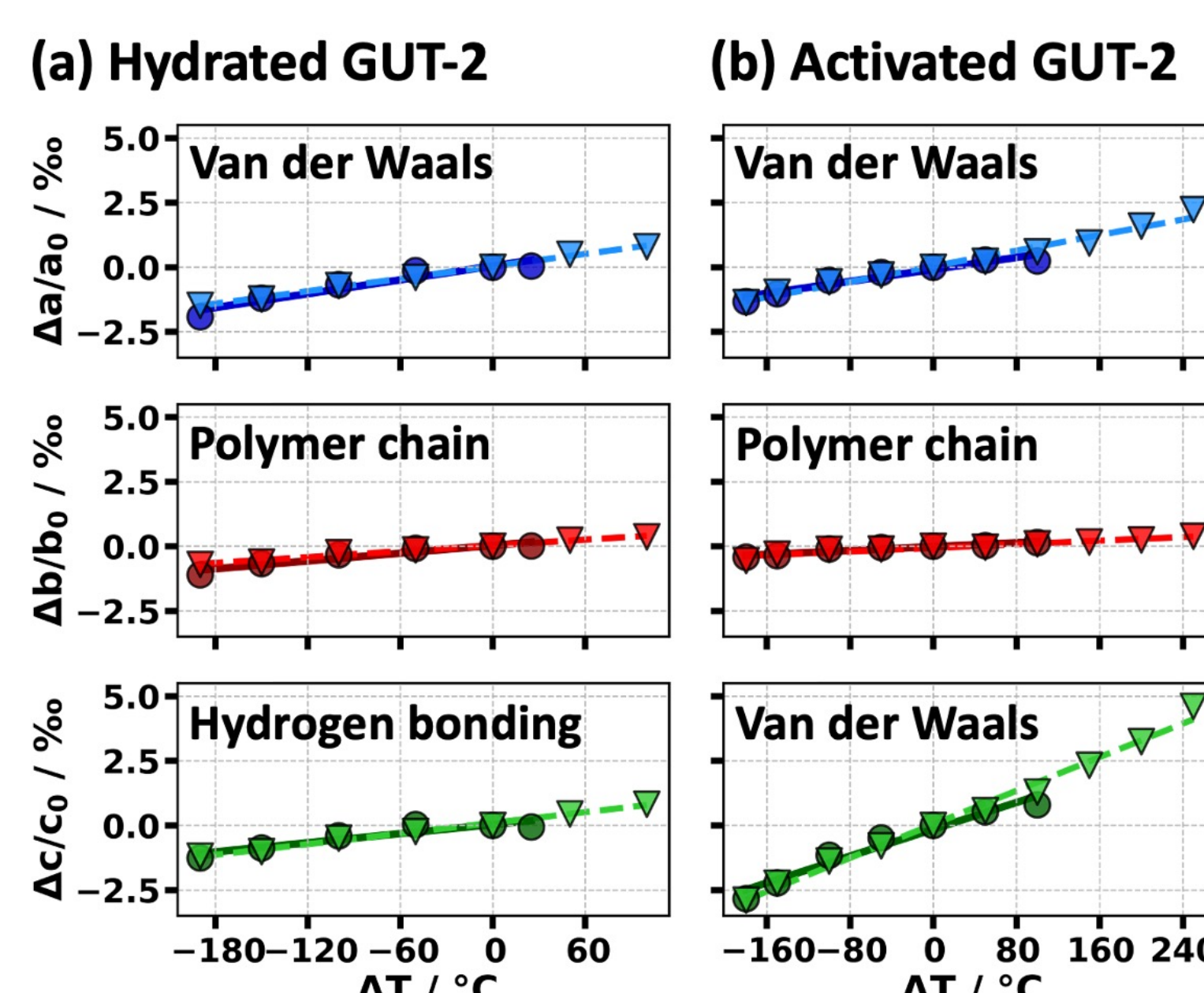
TEMPERATURE-DEPENDENT PXRD MEASUREMENTS



RIETVELD REFINEMENTS AND CRYSTALLIZATION KINETICS



ANISOTROPIC THERMAL EXPANSION COEFFICIENTS



Linear fit: $\alpha_L = \left(\frac{\partial L}{\partial T} \right)_p \frac{1}{L_0}$
with $L = \{a, b, c\}$

	α_a [10 ⁻⁶ K ⁻¹]	α_b [10 ⁻⁶ K ⁻¹]	α_c [10 ⁻⁶ K ⁻¹]
Hydrated GUT-2	8-9	1-3	6-7
Activated GUT-2	6-8	1-2	13-16

REFERENCES

- [1] R. Gross and A. Marx, *Festkörperphysik*, Walter de Gruyter, Berlin, 3rd edn., 2018.
- [2] K. Kodolitsch, A. Torvisco, T. Kamencek, M. Mazaj, E. Zojer, C. Slugovc, *Eur. J. Inorg. Chem.* 2025, e202500032.



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Related work:
Elastic constants of GUT-2 determined with Brillouin spectroscopy

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